

EC331 Research Proposal – Mahima Sabberwal

The paradox of significant improvement in education among women alongside persistently low female labour force participation (FLFP) rates in India has garnered several hypotheses, considering social norms and behaviours. I seek to investigate **how the level of education obtained by women in India affects their labour market participation and outcomes**. Better understanding the causes of low FLFP rates could lead to more informed interventions to raise participation, with improved welfare and efficiency at the micro and macro level (Hsieh, 2019; Qian, 2008).

Previous relevant work:

The U-shape relationship between female labour force participation and economic development is often cited to explain global trends in women's employment. Strong income and substitution effects drive this, as initially moving from household production and agriculture to the wider market results in higher household earnings and women withdrawing from wage work, until returns are high enough for the substitution effect to dominate. The decline after moderate education levels partially arises from social stigmas against women in manual work, the main occupation available outside of home-based activities (Goldin, 1994).

The trends between a woman's level of education and probability of employment in India is explored using 2012 data by a logit regression, controlling for personal and household characteristics (Chatterjee et al., 2018). More educated women often marry partners with higher levels of education, and into a household with more income. The authors find the income effect to likely partially explain the decline in FLFP in the U-shape, but highlight occupational sex segregation, where moderately educated women are excluded from clerical and sales work. Aligning with this, women appear to have greater difficulty matching to jobs when unemployed than men (Fletcher et al., 2017). Additionally, demand for white-collar workers has not matched growing supply, and more educated women tend to opt out of the workforce, rather than work in lower status jobs (Das & Zumbyte, 2017).

Method/identification strategy:

The existence of confounders, many likely unobservable, being correlated with both education and labour market decisions, creates significant endogeneity concerns in establishing causality and isolating effects. For example, a possible instrument to control for individual attributes, and to some extent characteristics of the household post-marriage, may be a sister's education level; but this would not account for social norms in early life affecting both education and employment decisions.

I will empirically consider the U-shape relationship between education and labour force participation for women, both nationally, and how the average education level for women by State corresponds to FLFP. I am currently directing my efforts to exploring how I may isolate the impacts of a constrained labour supply compared to labour demand factors (arising from few vacancies or discrimination by employers) contributing to the low FLFP rate, particularly for moderately educated women (at secondary school level).

To investigate the income effect, I seek to exploit an exogenous household income shock to compare changes in FLFP for affected and unaffected groups, utilising a differences-in-differences estimator. However, identifying a suitable shock for the demographic of interest (moderately educated women) has proven to be difficult. My initial idea to use investment-based income rises (if identifiable stock market boom with limited endogeneity) would not be appropriate due to the low likelihood of these families having such investments. Tax rebates and welfare payments in India also have elements of endogeneity, but it may be useful to explore these avenues and data availability further.

Data: I am reviewing literature to explore available data further (perhaps State specific), but am considering using:

- **Consumer Pyramids Household Survey:** Data on individuals' attributes, household wealth and investment decisions, income structure and consumption expenditure (thrice a year).
- **India Human Development Survey:** Panel data (2004-2005, 2011-2012), limited in timespan and recency. Includes personal characteristics, household structure, and employment data.
- **IZA International Data Science Centre:** data providing information on job seekers by gender, and vacancies on a low-wage online job portal across 2019 – 2020, by State and aggregate.
- Datasets available on **National Data and Analytics Platform (NDAP):** Includes data on crime against women by State. Could help explore/control for how safety perceptions affect FLFP.

Bonus research questions:

- 1) **How has the availability of free bus travel for women in Indian states affected average female labour force participation?** From 2019 - 2023, four states in India have implemented policies allowing women to travel for free on state-owned buses, varying in the time at which the schemes were launched. A Differences-in-Differences approach could be utilised to compare changes in the proportion of women formally employed before and after the introduction of free bus travel. I would need to consider pre-trends and whether any additional State-specific policies aiming to encourage women's employment are implemented in the time period examined. This analysis would also be narrowed to a particular socio-economic group who would be likely to frequent buses in India.
- 2) **How does the occurrence of a major crime against women in an Indian state affect female labour force participation, and how persistent are these effects?** Indian politicians acknowledge a link between women's safety and their low participation in the workforce, with this often being cited by families as a reason behind restricting and managing women's employment (Reed & Singh, 2024). The short- and long-term impacts of a major, publicly reported crime could be assessed through conducting event studies, but I would further explore alternative identification strategies.

References:

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